

Axiomatic 5D Spiral Topology

A Geometric–Cybernetic Framework

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Abstract

An application-neutral, axiomatic framework is introduced in which states are modeled as points in a five-dimensional space and structured trajectories appear in the form of spiral paths. The notion of a *5D spiral topology* is formalized as a geometric–cybernetic generalization of classical (in particular Cartesian) coordinate systems. The framework comprises: (i) a 5D state space with projection onto a three-dimensional “spatial” and a two-dimensional “internal” part, (ii) a distinguished class of spiral curves, (iii) resonance fields and action functionals, (iv) a field-driven feedback dynamics. Special classes of spiral paths (Archimedean and logarithmic spirals) are formally distinguished and embedded into the 5D context.

Keywords: 5D spiral topology; geometric–cybernetic framework; dynamical systems; resonance fields; high-dimensional state space.

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1 Introduction

Many systems with complex structure can be understood as dynamics in a high-dimensional state space. Classical coordinate systems typically rely on linear, orthogonal axes, whereas real-world processes often exhibit recursive, cyclic, and spiral patterns. The idea of a *spiral topology* is to reflect such patterns already at the level of the coordinate and trajectory structure.

This work introduces an application-independent framework in which a five-dimensional space is equipped with a distinguished class of spiral paths, resonance fields, and a cybernetic feedback process. The construction is deliberately neutral and does not refer to any particular domain; it is intended as a universal foundation upon which later specializations can be built.

The main goals are:

- to define an axiomatic notion of a 5D spiral topology,
- to introduce a spiral matrix as a discrete representation of spiral trajectories,
- to formally distinguish special classes of spiral paths (Archimedean vs. logarithmic),
- to describe resonance fields and feedback dynamics,
- to embed the construction into a geometric–cybernetic viewpoint.

2 Axiomatic Framework of a 5D Spiral Topology

2.1 Basic objects

Definition 2.1 (Spiral space structure). A *5D spiral space structure* is a tuple

$$\mathcal{S} = (X, \Gamma, \pi_{\text{spatial}}, \pi_{\text{internal}}, \Psi, \mathcal{W}, \Phi),$$

consisting of

- a nonempty set X of states,
- a set Γ of curves $\gamma : I \rightarrow X$ (spiral paths),
- projections $\pi_{\text{spatial}} : X \rightarrow \mathbb{R}^3$, $\pi_{\text{internal}} : X \rightarrow \mathbb{R}^2$,
- a family of resonance fields $\Psi = \{\psi, \rho, \omega\}$ with $\psi, \rho, \omega : X \rightarrow \mathbb{R}$,
- an action functional $\mathcal{W}$ on finite sequences in X ,
- a (discrete or continuous) dynamics operator Φ on X .

The following axioms specify when such a structure can be regarded as a *5D spiral topology*.

2.2 Axioms of a 5D spiral topology

Axiom 2.2 (Dimensionality and geometry). X is a real vector space of dimension 5 endowed with a Euclidean norm $\|\cdot\|$. There is a vector space isomorphism

$$X \cong \mathbb{R}^3 \times \mathbb{R}^2.$$

Axiom 2.3 (Projections). The maps

$$\pi_{\text{spatial}} : X \rightarrow \mathbb{R}^3, \quad \pi_{\text{internal}} : X \rightarrow \mathbb{R}^2$$

are linear and surjective, and satisfy

$$s = (\pi_{\text{spatial}}(s), \pi_{\text{internal}}(s)) \quad \text{for all } s \in X.$$

Axiom 2.4 (Spiral paths). Γ is a nonempty subset of the set of continuously differentiable curves $\gamma : I \rightarrow X$ with $I \subset \mathbb{R}$ an interval, such that for every $\gamma \in \Gamma$:

1. The projection $\pi_{\text{spatial}} \circ \gamma$ can be described in cylindrical coordinates (r, φ, h) .
2. $r(t) \geq 0$ is defined for all t and is monotone on I (e.g. non-decreasing).
3. $\varphi(t)$ is strictly monotone on I .

Such curves are called *spiral paths*.

Axiom 2.5 (Resonance fields). The resonance fields

$$\psi, \rho, \omega : X \rightarrow \mathbb{R}$$

are at least once continuously differentiable.

Axiom 2.6 (Action functional). *The action functional*

$$\mathcal{W} : \bigcup_{n \in \mathbb{N}} X^n \rightarrow \mathbb{R}$$

assigns a real value to each finite sequence $(s_1, \dots, s_n)$ and is additive under concatenation:

$$\mathcal{W}(s_1, \dots, s_k, s_{k+1}, \dots, s_n) = \mathcal{W}(s_1, \dots, s_k) + \mathcal{W}(s_{k+1}, \dots, s_n)$$

for all $1 \leq k < n$.

Axiom 2.7 (Dynamics operator). *Φ is either*

- *a map $\Phi : X \rightarrow X$ (discrete time), or*
- *a vector field $\Phi : X \rightarrow X$ (continuous time),*

such that for each $\gamma \in \Gamma$, the curve

$$\tilde{\gamma}(t) = \Phi(\gamma(t))$$

belongs again to Γ , provided γ respects the dynamics (i.e. γ is an orbit of an iteration or flow generated by Φ).

Definition 2.8 (5D spiral topology). A spiral space structure $\mathcal{S} = (X, \Gamma, \pi_{\text{spatial}}, \pi_{\text{internal}}, \Psi, \mathcal{W}, \Phi)$ satisfying Axioms 2.2–2.7 is called a *5D spiral topology*.

3 5D State Space and Coordinate Structure

3.1 States and projections

Under Axiom 2.2 we may identify X explicitly with $\mathbb{R}^5$.

Definition 3.1 (5D state space). Set

$$X := \mathbb{R}^5.$$

A state is a vector

$$s = (x_1, x_2, x_3, x_4, x_5)^\top \in X.$$

Definition 3.2 (Projection maps). In accordance with Axiom 2.3 we define

$$\pi_{\text{spatial}} : X \rightarrow \mathbb{R}^3, \quad \pi_{\text{spatial}}(s) = (x_1, x_2, x_3), \quad (3.1)$$

$$\pi_{\text{internal}} : X \rightarrow \mathbb{R}^2, \quad \pi_{\text{internal}}(s) = (x_4, x_5). \quad (3.2)$$

3.2 Cylindrical representation of the spatial projection

Definition 3.3 (Cylindrical coordinates). For $s \in X$ with $\pi_{\text{spatial}}(s) = (x, y, z)$ let

$$r = \sqrt{x^2 + y^2}, \quad (3.3)$$

$$\varphi = \text{atan2}(y, x), \quad (3.4)$$

$$h = z. \quad (3.5)$$

Then the spatial component is given in cylindrical coordinates as (r, φ, h) .

Remark 3.4. The full 5D state can be written as

$$s = (r \cos \varphi, r \sin \varphi, h, x_4, x_5).$$

This suggests an interpretation as a 3D spiral with two internal coordinates, but the framework itself does not impose any specific semantic meaning.

4 Spiral Paths, Spiral Sequences, and Spiral Matrix

4.1 Spiral paths

Definition 4.1 (Spiral path). Let $I \subset \mathbb{R}$ be an interval. A smooth curve

$$\gamma : I \rightarrow X$$

is called a *spiral path* if it satisfies the conditions of Axiom 2.4, i.e. its spatial projection admits a description in cylindrical coordinates (r, φ, h) with r monotone and φ strictly monotone.

4.2 Discrete samples and spiral matrix

Definition 4.2 (Spiral sequence). Let $\gamma : I \rightarrow X$ be a spiral path and $(t_i)_{i=1}^n$ a strictly increasing sequence in I . The sequence

$$S_i := \gamma(t_i) \in X, \quad i = 1, \dots, n,$$

is called a *spiral sequence*.

Definition 4.3 (Spiral matrix). The *spiral matrix* associated with a spiral sequence $(S_i)_{i=1}^n$ is

$$\mathbf{S} = \begin{pmatrix} S_1^\top \\ S_2^\top \\ \vdots \\ S_n^\top \end{pmatrix} \in \mathbb{R}^{n \times 5}.$$

Remark 4.4. The spiral matrix can be viewed as a high-dimensional coordinate list in which the ordering of rows is determined by the spiral trajectory.

Definition 4.5 (Spiral graph). The *spiral graph* associated with a spiral sequence $(S_i)_{i=1}^n$ is the directed graph with

- vertex set $V = \{S_1, \dots, S_n\}$,
- edge set $E = \{(S_i, S_{i+1}) : i = 1, \dots, n-1\}$.

5 Special Classes of Spiral Paths

5.1 Archimedean spirals in 5D

Definition 5.1 (Archimedean 5D spiral). Let $a, b, c, \alpha, \beta \in \mathbb{R}$ with $b > 0$. A curve $\gamma : [0, \infty) \rightarrow X$ is called an *Archimedean 5D spiral* if it is given by

$$r(t) = a + bt, \quad (5.1)$$

$$\varphi(t) = t, \quad (5.2)$$

$$h(t) = ct, \quad (5.3)$$

$$x_4(t) = \alpha t, \quad (5.4)$$

$$x_5(t) = \beta t \quad (5.5)$$

and

$$\gamma(t) = (r(t) \cos \varphi(t), r(t) \sin \varphi(t), h(t), x_4(t), x_5(t)).$$

Proposition 5.2. *For an Archimedean 5D spiral, the radial distance between successive turns satisfies*

$$\Delta r := r(t + 2\pi) - r(t) = 2\pi b,$$

i.e. the radial spacing of turns is constant.

Proof. Using $r(t) = a + bt$ we have

$$r(t + 2\pi) - r(t) = (a + b(t + 2\pi)) - (a + bt) = 2\pi b.$$

□

5.2 Logarithmic spirals in 5D

Definition 5.3 (Logarithmic 5D spiral). Let $a > 0$ and $b, c, \alpha, \beta \in \mathbb{R}$. A curve $\gamma : \mathbb{R} \rightarrow X$ is called a *logarithmic 5D spiral* if

$$r(t) = ae^{bt}, \quad (5.6)$$

$$\varphi(t) = t, \quad (5.7)$$

$$h(t) = ct, \quad (5.8)$$

$$x_4(t) = \alpha t, \quad (5.9)$$

$$x_5(t) = \beta t \quad (5.10)$$

and

$$\gamma(t) = (r(t) \cos \varphi(t), r(t) \sin \varphi(t), h(t), x_4(t), x_5(t)).$$

Proposition 5.4. *For a logarithmic 5D spiral one has*

$$r(t + 2\pi) = e^{2\pi b} r(t),$$

i.e. the radius is multiplied by the constant factor $e^{2\pi b}$ after each full turn.

Proof. Direct computation yields

$$r(t + 2\pi) = ae^{b(t+2\pi)} = ae^{bt} e^{2\pi b} = e^{2\pi b} r(t).$$

□

5.3 Generalized 5D spirals

Definition 5.5 (Generalized 5D spiral). A curve $\gamma : I \rightarrow X$ is called a *generalized 5D spiral* if it has the form

$$\gamma(t) = (R(t) \cos \Phi(t), R(t) \sin \Phi(t), H(t), U(t), V(t))$$

with smooth functions $R, \Phi, H, U, V : I \rightarrow \mathbb{R}$, and

- $R(t) \geq 0$ for all t and R is monotone on I ,
- Φ is strictly monotone on I .

Proposition 5.6. *Archimedean and logarithmic 5D spirals are special cases of generalized 5D spirals with*

$$R(t) = a + bt \quad \text{or} \quad R(t) = ae^{bt}, \quad \Phi(t) = t.$$

6 Resonance Fields and Weights

6.1 Scalar fields

Definition 6.1 (Resonance fields). The fields

$$\psi, \rho, \omega : X \rightarrow \mathbb{R}$$

are called *resonance fields*. They can be interpreted abstractly as intensity, coherence, and rhythm fields, without presupposing any concrete application domain.

6.2 Weights of spiral segments

Definition 6.2 (Spiral segment). Given a spiral sequence $(S_i)_{i=1}^n$, a *spiral segment* is a pair

$$K_i := (S_i, S_{i+1}) \in X \times X, \quad i = 1, \dots, n-1.$$

Definition 6.3 (Segment weight). Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ be smooth. The weight of a spiral segment $K_i = (S_i, S_{i+1})$ is defined as

$$w(K_i) := f(\psi(S_i), \rho(S_i), \omega(S_i)).$$

Definition 6.4 (Spiral action functional). For a spiral sequence $(S_i)_{i=1}^n$ define

$$\mathcal{W}((S_i)_{i=1}^n) := \sum_{i=1}^{n-1} w(K_i).$$

6.3 Resonance distance and resonance criterion

Definition 6.5 (Resonance distance). For $s, s' \in X$ and weights $w_\psi, w_\rho, w_\omega \geq 0$ define

$$d_R(s, s') := w_\psi |\psi(s) - \psi(s')| + w_\rho |\rho(s) - \rho(s')| + w_\omega |\omega(s) - \omega(s')|.$$

Definition 6.6 (Resonance criterion). For $\varepsilon > 0$, a state s is called *resonant* to s' if

$$d_R(s, s') < \varepsilon,$$

and *divergent* from s' if $d_R(s, s') \geq \varepsilon$.

7 Cybernetic Feedback in Spiral Space

7.1 Discrete dynamics

Definition 7.1 (Spiral dynamical system (discrete time)). A *spiral dynamical system* in discrete time is a sequence $(S_k)_{k \in \mathbb{N}}$ in X with

$$S_{k+1} = \Phi(S_k),$$

where $\Phi : X \rightarrow X$ is the dynamics operator.

Example 7.2 (Gradient-based feedback). Let $\lambda_\psi, \lambda_\rho, \lambda_\omega \in \mathbb{R}$ and set

$$F(s) := \lambda_\psi \nabla \psi(s) + \lambda_\rho \nabla \rho(s) + \lambda_\omega \nabla \omega(s).$$

Define

$$\Phi(s) := s + \alpha F(s), \quad \alpha > 0.$$

Then the iteration

$$S_{k+1} = S_k + \alpha (\lambda_\psi \nabla \psi(S_k) + \lambda_\rho \nabla \rho(S_k) + \lambda_\omega \nabla \omega(S_k))$$

fulfills the form required by Axiom 2.7.

7.2 Branching under resonance criteria

Definition 7.3 (Spiral branching). Let $(S_k)_{k \in \mathbb{N}}$ be a spiral dynamical system and $S_{\text{ref}} \in X$ a reference state. If for some k the condition

$$d_R(S_k, S_{\text{ref}}) \geq \varepsilon$$

holds, a *branching* at step k may be introduced: one constructs an alternative sequence $(\tilde{S}_j)_{j \geq k}$ with

$$\tilde{S}_k = S_k,$$

which follows a modified dynamics $\tilde{\Phi}$, e.g. with different parameters in F .

8 5D Spiral Topology and Coordinate Systems

8.1 Comparison with Cartesian systems

In the Cartesian setting, points are described by independent coordinates $(x_1, \dots, x_n)$ and the topology is induced by the standard norm. In a 5D spiral topology, there is an additional trajectory-based ordering:

- Points are characterized not only by their position but also by their embedding in spiral paths.
- Resonance fields induce a modified metric structure, extending the pure Euclidean geometry.

Proposition 8.1. *Let $\gamma : I \rightarrow X$ be a spiral path with strictly monotone radius $R(t)$ and angle $\Phi(t)$ in its spatial projection. Then the map*

$$\Phi_{\text{spatial}} : I \rightarrow \mathbb{R}^3, \quad t \mapsto \pi_{\text{spatial}}(\gamma(t))$$

is injective if self-intersections of the curve are excluded.

Proof. Assume there exist $t_1 \neq t_2$ such that $\Phi_{\text{spatial}}(t_1) = \Phi_{\text{spatial}}(t_2)$. In cylindrical coordinates this implies $r(t_1) = r(t_2)$, $h(t_1) = h(t_2)$, and $\Phi(t_1) = \Phi(t_2) + 2\pi k$ for some $k \in \mathbb{Z}$. Strict monotonicity of R and Φ rules out such equalities unless the curve has self-intersections, contrary to the assumption. $\square$

8.2 Spiral coordinate assignment to discrete sets

Definition 8.2 (Spiral coordinate transformation). Let $I = \{1, \dots, n\}$ be an index set and $E : I \rightarrow X$ be injective. A *spiral coordinate assignment* of I is a choice of ordering of I such that the sequence $(E(i))_{i \in I}$ forms a spiral sequence.

Remark 8.3. In contrast to a Cartesian grid, where indices are arranged in a rectangular lattice, indices are here organized along spiral paths. The spiral matrix $\mathbf{S}$ is the explicit representation of this assignment.

9 Discussion and Outlook

An application-neutral, axiomatic framework for a 5D spiral topology has been developed. The essential components are:

- a five-dimensional state space with projections onto spatial and internal coordinates,
- a distinguished class of spiral paths and their discrete representation via spiral sequences and spiral matrices,
- resonance fields and an action functional for path evaluation,
- a dynamics operator describing cybernetic feedback,
- the formal distinction between Archimedean, logarithmic, and generalized spiral paths.

The framework is fully neutral and can be instantiated in various application domains without being tied to any particular one. Potential directions for further work include:

- detailed stability analysis of spiral dynamical systems,
- extension by alternative metrics combining Euclidean and resonance-based components,
- investigation of spaces of spiral paths using topological or measure-theoretic methods,
- algebraic and category-theoretic descriptions of branching and composition of spirals.

The result is a self-contained, universal model for the geometry and cybernetics of spiral structures in a 5D space, which can serve as a basis for a broad range of specialized constructions.